POTPRESS · FEASIBILITY

PotPress feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Working force limit 5.4 t (D9, decided by Amish 2026-10-25)
0.2	2026-10-23	Amish Chadha	Updated for the STEM check fixes (PPR-DDR-006)
0.1	2026-10-23	Amish Chadha	First assessment (kit 1.8 upgrade)

PotPress feasibility assessment

PotPress is feasible on paper: the structure, hydraulics and guarding pass their checks, and the open items are process knowledge (forming force, clay shrinkage) and local supply (a foundry, a true 250 mm printer) that only a partner factory can settle. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Frame, pin and bolts carry 1.5 times the 20 t jack without yield; 0.66 mm deflection at 10 t; molds close on a metal stop	High	Forming force unknown until trials (decided 2026-10-02), capped at 5.4 t by the handle (D9, 2026-10-25); clay shrinkage now in the mold size at an assumed 8 %, to be measured (PPR-DDR-006)	PPR-STM-001, PPR-CAL-001
Materials	S275 channel and plate, grade 10.9 bolts, cast aluminium molds, polyethylene liners	Medium	A356 wheel scrap sorted and each melt checked (PPR-DDR-006); foundry OES or XRF availability to assess; casting porosity on mold faces	PPR-CAL-001, BOM
Manufacturing and assembly	Stick welding, drilling and bolting; 16 printed pattern segments; molds hand-finished to templates	Medium	Wall evenness depends on hand finishing (PR-002 at risk); patterns need a true 250 x 250 mm bed (PR-007 at risk); parts over 25 kg go on a chain block (PPR-DDR-006)	PPR-BLD-001
Power and controls	Hand-pumped hydraulics; no mains power; mechanical interlocks	High	Working force limited to 5.4 t by the 150 N handle limit with 85 % pump efficiency (D9, decided 2026-10-25); more needs a separate pump (D6, open)	PPR-STM-001 S8
Supply of parts	Steel, bottle jacks and hardware in most regional towns; aluminium casting may need a city foundry	Medium	Foundry access is the first partner question (PRB open questions)	BOM, PPR-PRB-001
Cost	Priced BOM $1,279 against the $1,060 value-engineering target	Medium	$219 over; the lifting set ($66) can be hired; savings listed in PPR-DEC-001	BOM, PPR-PRS-001 PR-010

Ratings: High, Medium, Low.